MULTIVARIABLE CALCULAS

1] **Limit of a fn**: we say that $f(x,y)$ approaches $l \in R$ as $(x,y) \to (x_0,y_0)$ if for given $\varepsilon > 0$, $\exists$ a $\delta > 0$ ($\delta(\varepsilon)$):

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta \Rightarrow |f(x,y) - l| < \varepsilon$$

$|x - x_0| < \delta$
$|y - y_0| < \delta$

2nd method to show limit DNE.

① To show that limit does not exist, it is enough to show that l is diff in 2 diff paths.

② But to show that it exists, use ε-δ definition only.

③ If limit Exist, $\lim_{x\to x_0}\left[\lim_{y\to y_0} f(x,y)\right] = \lim_{y\to y_0}\left[\lim_{x\to x_0} f(x,y)\right]$

If its not equal then limit DNE, but if its equal, limit may or may not exist.

Best method to show that limit DNE.

④ If P/q are polynomials of x, y, then if $\deg(P) \leq \deg(q)$ then limit D.N.E.

2] **Continuity**: limit must be equal to fn value at that pt.

3] **Partial Derivatives**: $\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = f_x(x_0,y_0) = \lim_{h\to 0} \frac{f(x_0+h, y_0) - f(x_0,y_0)}{h}$

4] **Higher Order Derivative**: $\left.\frac{\partial^2 f}{\partial x^2}\right|_{(x_0,y_0)} = f_{xx}(x_0,y_0) = \lim_{h\to 0} \frac{f_x(x_0+h,y_0) - f_x(x_0,y_0)}{h}$

ex- $\left.\frac{\partial^3 f}{\partial x^2 \partial y}\right|_{(x_0,y_0)} = \lim_{h\to 0} \frac{f_{xy}(x_0+h, y_0) - f_{xy}(x_0,y_0)}{h}$

$\boxed{\# \lim_{h\to 0} \frac{h}{|h|} = \text{D.NE}}$ → Common Sense. :)

5] **Geometrical Interpretation of Partial Derivative**.

$$z - f(x_0,y_0) = f_x(x_0,y_0).(x - x_0)$$

$y - y_0 = m(x - x_0)$

6) **Differentiability**: Diff if

$$f(x_0+h) - f(x_0) = h.\alpha + h.\phi(h)$$

$\alpha = f'(x_0)$

Here α, β are partial derivatives wrt ~~x~~ x, y.

$$f(x_0+h, y_0+k) - f(x_0,y_0) = (h,k).(\alpha,\beta) + (h,k)(\phi,k)$$

~~Do not use~~ $\frac{\partial f}{\partial x} = x^4 + 3x^2y$] Matlab direct differentiation to calculate $\left.\frac{\partial f}{\partial x}\right|_{(0,0)}$

Agar fn diya hai, with definition {, then for $(x,y) \neq (0,0)$ mein direct diff karna hai, & $(x,y) = (0,0)$ mein h,k wala method use karna hai.

$$f(x_0+h, y_0+k) - f(x_0,y_0) = h.\alpha + k.\beta + h\,\phi(h,k) + k\,\psi(h,k)$$

7] **Necessary Condition of Diff**

$f_x(x_0,y_0)$ and $f_y(x_0,y_0)$ must exist

But if it exist then fn may or may not be diff.

ye dono ϕ, ψ pe fn ki value should exist

8] **Chain Rule**: $f \langle^{x}_{y} \rangle t$.

$$\frac{df}{dt} = \frac{\partial f}{\partial x}.\frac{dx}{dt} + \frac{\partial f}{\partial y}.\frac{dy}{dt}$$

$f \langle^{x}_{y} \rangle s,t$.

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}.\frac{\partial y}{\partial s}$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}.\frac{\partial y}{\partial t}$$

9] **Implicit functions**: $\frac{\partial y}{\partial x} = \frac{-\frac{\partial F}{\partial x}}{+\frac{\partial F}{\partial y}}$

$\boxed{\text{If } \frac{\partial^2 f}{\partial x \partial y} \text{ \& } \frac{\partial^2 f}{\partial y \partial x} \text{ are continuous, then they are equal}}$

If f is diff then total derivative is

$$df = \frac{\partial f}{\partial x}.dx + \frac{\partial f}{\partial y}.dy$$ ~~is total~~

9] **Homogenous fn**: ① $f(\lambda x, \lambda y) = \lambda^n f(x,y)$

② $f(x,y) = x^n g\left(\frac{y}{x}\right)$

③ $f(x,y) = y^n g\left(\frac{x}{y}\right)$

degree.

10] **Euler's Thm of Homogenous fn**:

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = n.f$$

$$x^2\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x \partial y} + y^2\frac{\partial^2 f}{\partial y^2} = n(n-1)f.$$

11] **Tangent Vector**: $\left(\frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j}\right).(dx\hat{i} + dy\hat{j}) = 0$

$\nabla f.T = 0$

Slope of tangent $= \frac{dy}{dx}$

$y - y_0 = \frac{dy}{dx}(x - x_0)$ → Tangent

Eqn of normal:

$$\frac{x - x_0}{\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)}} = \frac{y - y_0}{\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)}}$$

12] Tangent plane to surface:

$$\left(i\frac{\partial f}{\partial x}+j\frac{\partial f}{\partial y}+k\frac{\partial f}{\partial z}\right)\cdot(i\,dx+j\,dy+k\,dz)=0$$

$$\nabla f \cdot T=0$$

Eqn of plane passing through point (x_0,y_0,z_0) with normal $\nabla f|_P$ is given by:

$$\frac{\partial f}{\partial x}\Big|_P (x-x_0)+\frac{\partial f}{\partial y}\Big|_P (y-y_0)+\frac{\partial f}{\partial z}\Big|_P (z-z_0)=0.$$

$$z=z_0+\frac{\partial f}{\partial x}\Big|_P (x-x_0)+\frac{\partial f}{\partial y}\Big|_P (y-y_0)$$

$$z_0=f(x_0,y_0)$$

eqn of normal

$$\frac{x-x_0}{\frac{\partial f}{\partial x}\big|_P}=\frac{y-y_0}{\frac{\partial f}{\partial y}\big|_P}=\frac{z-z_0}{\frac{\partial f}{\partial x}\big|_P}$$

13] Taylor's Thm:

$$f(x,y)=f(x_0,y_0)+\frac{1}{1!}\left[(x-x_0)\frac{\partial}{\partial x}+(y-y_0)\frac{\partial}{\partial y}\right]f(x_0,y_0)$$
$$+\frac{1}{2!}\left[(x-x_0)\frac{\partial}{\partial x}+(y-y_0)\frac{\partial}{\partial y}\right]^2 f(x_0,y_0)\dots$$
$$\dots\frac{1}{n!}\left[(x-x_0)\frac{\partial}{\partial x}+(y-y_0)\frac{\partial}{\partial y}\right]^n f(x_0,y_0)+R_n$$

$$R_n=\frac{1}{(n+1)!}\left[(x-x_0)\frac{\partial}{\partial x}+(y-y_0)\frac{\partial}{\partial y}\right]^{n+1} f((1-\theta)x_0+\theta x,\ (1-\theta)y_0+\theta y)$$

$0<\theta<1$

14] Error in the linear approximation:

$$|R_1|\le\frac{1}{2}(\delta_1+\delta_2)^2.B$$

$B=\max\{|f_{xx}|,|f_{xy}|,|f_{yy}|\}$

$|x-x_0|<\delta_1$

$|y-y_0|<\delta_2$

15] Error in Quadratic Approximation: $\frac{B}{6}\{(\delta_1+\delta_2)^3\}$

$B=\max\{|f_{xxx}|,|f_{xxy}|,|f_{xyy}|,|f_{yyy}|\}$

$|x-x_0|<\delta_1$; $|y-y_0|<\delta_2$

16] Maclaurin's Thm:

$$f(x,y)=f(0,0)+\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)f(0,0)+\frac{1}{2}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^2 f(0,0)+\dots+\frac{1}{n!}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^n f(0,0)+R_n$$

$$R_n=\frac{1}{(n+1)!}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^{n+1} f(\theta x,\theta y)$$

$0<\theta<1$

16] Laplace's Eqn: $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$

17] Heat Eqn: ① $\frac{\partial u}{\partial t}=k\frac{\partial^2 u}{\partial x^2}$ ② $\frac{\partial^2 u}{\partial t^2}=k\frac{\partial^2 u}{\partial x^2}$

18] Condition for existance of extremum:

Continuous, first order derivative exist, then $f_x(x_0,y_0)=0$

$f_y(x_0,y_0)=0.$

Sufficient conditions: ① $rt-s^2>0$; $r>0$ → min.

② $rt-s^2>0$; $r<0$ → max.

$r=f_{xx}(x_0,y_0)$; $s=f_{xy}(x_0,y_0)$; $t=f_{yy}(x_0,y_0)$

③ If $rt-s^2<0$ → saddle pt.

④ If $rt-s^2=0$ → No conclusion.

INTEGRAL CALCULAS

1] $\iint_D f(x,y)\,dA$: Volm of solid bounded by $z = f(x,y)$ & D.

2] Limits identification: function wala limit andar, const limit bahar.

When limits are directly given in question, use it, But if its given in form of some shape or vertices of that shape, then make functional limits using vertices.

Note: Necessary condition for existance of $\iint_D f(x,y)\,dA$ is boundness of $f(x,y)$ over D. That means, if $f(x,y)$ is not bounded over D, then $\iint_D f(x,y)$ DNE.

3] Fubini's Theorem: If function is continuous then

$$\iint_D f(x,y)\,dx\,dy = \iint_D f(x,y)\,dy\,dx$$

If not equal, then fn is not continuous.

* Volm of Ellipsoid = $\frac{4}{3}\pi abc$ Because Area = πab.

4] Area by double Integration: $\iint_D dx\,dy$ when $f(x,y) = 1$.

length of Curve: $\int_D dx$ When $f(x) = 1$.

5] Change in variables:

$$\iint_D f(x,y)\,dx.dy = \iint_{D^*} f(x(u,v),y(u,v))\,|J(x,y)|\,du.dv.$$

$$J(x,y) = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

- Polar: $\iint_D f(x,y)\,dx\,dy = \iint_{D^*} f(r\cos\theta, r\sin\theta).\,r\,dr\,d\theta$

Be carefull while choosing limits.

6] Physical Applications: ① Mass: where Density: $f(x,y)$.

$$M = \iint_D f(x,y)\,dx\,dy.$$

② Centre of gravity: $\bar{x} = \frac{1}{M}\iint_D x.f(x,y)\,dx\,dy.$

$\bar{y} = \frac{1}{M}\iint_D y\,f(x,y)\,dx\,dy.$

Iske liye upar ke formula se mass nikalna padta hai.

③ Moment of Inertia: $I_x = \iint_D y^2 f(x,y)\,dx\,dy$

$I_y = \iint_D x^2 f(x,y)\,dx\,dy.$

$I_o = I_x + I_y \rightarrow$ moment of inertia of the mass about origin.

$I_x = \iint (y-b)^2 f(x,y)\,dx\,dy$; $I_y = \iint (x-a)^2 f(x,y)\,dx\,dy$

about lines $x = a$, $y = b$.

4] Average value: $\frac{1}{A}\iint_D f(x,y)\,dx\,dy$.

Area = (A)

Volume between 2 surfaces [Imp]

① Ek fn x,y ke terms mein hoga. use polar substitution to simplify it.

② Find Jacobian.

③ Make rough fig of x,y wala function.

④ Udhar se θ ke limits milte hai.

⑤ ~~Find θ ke limits from equations as well~~

⑥ Then ~~find~~ do the integration.

If x,y wala seperate function nhi diya toh, ~~make z = 0 & get a fn in form of~~ x,y.

Tip- har baar agar lg rha hai ki x,y mein solve karna mushkil hai, just convert into u,v & r,θ.

7] Triple Integrals: $\iiint_E f(x,y,z)\,\underbrace{dx\,dy\,dz}_{V}$

If limits are not given, especially for constant thing, find it using graph. ex-Pg $\boxed{47}$

8] Application of Triple Integrals:

① Hyper Volume: $\iiint_E f(x,y,z)\,dx\,dy\,dz.$

② Volume: $\iiint_E dx\,dy\,dz$

To find Volume inside some solid, find limits of z, y, x, & do this. Can be done using double integral as well.

9] Change of variables in triple integrals:

$$\iiint_E f(x,y,z)\,dx\,dy\,dz = \iiint_{E^*} |J|\, f(x(u,v,w), y(u,v,w), z(u,v,w))\,du\,dv\,dw.$$

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)}$$

~~Spherical Coordina~~ Cartesian.

10] Cylindrical Coordinates:

$x = r\cos\theta,\ y = r\sin\theta,\ z = z$

$r^2 = x^2 + y^2$

$$J = \frac{\partial(x,y,z)}{\partial(r,\theta,z)} = r$$

$$\iiint_E f(x,y,z)\,dx\,dy\,dz = \iiint_{E^*} f(x(r,\theta,z), y(r,\theta,z), z(r,\theta,z))\,|r|\,dr\,d\theta\,dz.$$

11] Spherical coordinates:

$x = r\cos\theta\sin\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\phi$

$r^2 = x^2+y^2+z^2$

$\theta \to$ angle b/w +ve x-axis & $\vec{OP}$

$\phi \to$ angle b/w +ve z-axis & $\vec{OP}$

Mostly: $0 \le \theta \le 2\pi$; $0 \le \phi \le \pi$

when +ve, +ve. $0 \le \theta \le \pi/2$; $0 \le \phi \le \pi/4$.

$$J = \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = r^2\sin\phi$$

To find the limits of r; find intersection of cone & plane. Then put polar coordinates in that. then you get the interval of r.

12] Physical Applications of Double Integrals:

① Mass: $M = \iiint_E f(x,y,z)\,dx\,dy\,dz$ ↳ Density.

② $\bar{x}$: $\frac{1}{m}\iiint_E x\, f(x,y,z)\,dx\,dy\,dz$

$\bar{y}$: $\frac{1}{m}\iiint_E y\, f(x,y,z)\,dx\,dy\,dz$

$\bar{z}$: $\frac{1}{M}\iiint_E z\, f(x,y,z)\,dx\,dy\,dz$.

} centre of Mass

③ I_x: $\iiint_E (y^2+x^2)\, f(x,y,z)\,dx\,dy\,dz$

I_y: $\iiint_E (x^2+z^2)\, f(x,y,z)\,dx\,dy\,dz$

I_z: $\iiint_E (x^2+y^2)\, f(x,y,z)\,dx\,dy\,dz$.

13] Drinchlet Integral:

$$I = \iiint_V x^{l-1} y^{m-1} z^{n-1}\,dx\,dy\,dz = \frac{\Gamma l\,\Gamma m\,\Gamma n}{\Gamma(l+m+n+1)}$$

14]

$$\iiint f(x,y,z)\, x^{l-1} y^{m-1} z^{n-1}\,dx\,dy\,dz = \frac{\Gamma l\,\Gamma m\,\Gamma n}{\Gamma(l+m+n)} \int_{h_1}^{h_2} f(u)\, u^{l+m+n-1}\,du.$$

* Ordinary Differential Equation

1] **ODE**: DE which involves one independent variable

2] **Partial DE**: DE which involves more than one independent variable & partial derivatives of dependent wrt independent.

3] **Order**: Highest order derivative.

4] **Degree**: Derivative ka highest power, after the eqn has been made free from radicals & fractions in derivatives.

5] **LDE**: ① Saare derivative terms are in 1^{st} degree.
② Derivative terms should not be in product with dependent variable.

$$y^{(n)} + c_1 y^{(n-1)} + c_2 y^{(n-2)} + \dots + c_n y = r(x) \quad *$$

where $c_1, c_2, c_3 \dots$ are constants or fn of x.

6] **Non Linear**: Which are not linear 😑

ex- $y\frac{dy}{dx} + \frac{d^3y}{dx^2} = \sin x + \cos x$.

It's like derivative chhodke kahi pe y^n hona nhi chahiye.
$n=1$ chalega.

7] **Formation of D.E**: If eqn contains n number of arbitrary constants, then order will be n.

8] **Solution of a D.E**: A function ~~is~~ $y=f(x)$ is called a solution of differential eqn on (a,b) if substitution of $y(x)$ & its derivatives satisfies the D.E.

*The General Solution of DE of order n contains n arbitrary constants.

9] **Exact differential E**: $df = M(x,y)\,dx + N(x,y)\,dy = 0$
partial derivative of f.

How to check? $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ → This shows that there exist a fn $f(x,y)$ such that $df = Mdx + Ndy = 0$.

How to find soln: i] Integrate M wrt x to get $f(x,y) = \int M dx + g(y)$
ii] Diff it wrt y & compare with N to get $\partial g/\partial y$.
iii] Integrate wrt to y to get $g(y)$ with arb. const.
iv] You got the soln. ü

- **Implicit** - $y = f(x)$ can be expressed
- **Explicit** - Cant be expressed independently.

10] **I.F**: If $Mdx + Ndy = 0$ is not exact but $\phi(x,y)Mdx + N\phi(x,y)dy = 0$ is exact then $\phi(x,y)$ is I.F.

Result-1 If $\dfrac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is the fn of x, $\to f(x)$
then $\boxed{e^{\int f(x)\,dx}}$ is the I.F

Result-2 If $\dfrac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = g(y)$, then ~~$e^{\int g(y)}$~~ $\boxed{e^{\int g(y)\,dy}}$ is IF

Result 3: If M & N are homogenous & $Mx + Ny \neq 0$ then $\frac{1}{Mx+Ny}$ is integrating factor.
But if $Mx + Ny = 0$ then $\frac{1}{xy} / \frac{1}{x^2} / \frac{1}{y^2}$ is IF.

- DE ka solution hamesha $f(x,y) = c$ mein likhna

- Har baar lamba method use karne se accha look if directly kisike derivatives diye hai.

11] **Linear First Order ODEs**: $\frac{dy}{dx} + P(x).y = r(x)$

then IF is $e^{\int p(x)\,dx}$

Soln → $y.e^{\int p(x)\,dx} = \int r(x).e^{\int p(x)dx} + C$.

12] **Bernoulli's Eqn**:

$$\frac{dy}{dx} + p(x).y = q(x).y^n$$

Procedure - i] divide by y^n RHS & LHS.
ii] Substitute $\frac{1}{y^{n-1}}$ at t &
iii] It will be converted to LDE.

13] **LDE solution by Inspection**: ① $\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$

② $\frac{xdy - ydx}{y^2} = -d\left(\frac{x}{y}\right)$

③ $\frac{xdy - ydx}{xy} = d\left(\log\left(\frac{y}{x}\right)\right)$

④ $\frac{xdy - ydx}{x^2+y^2} = d\left(\tan^{-1}\left(\frac{y}{x}\right)\right) = -d\left(\tan^{-1}\left(\frac{x}{y}\right)\right)$

⑤ $\frac{xdx + ydy}{x^2+y^2} = \frac{1}{2}\, d\left(\log(x^2+y^2)\right)$

14] **Orthogonal Trajectories**: family of curves intersecting each other ⊥ly.
Procedure - Replace y' by $-1/y'$ and integrate back.

Polar: $f(r,\theta,c) \xrightarrow{\text{diff}} \phi(r,\theta,r') = 0$

Now replace r' by $-r^2/r'$

15] **Existence Theorem**: If $f(x,y)$ is cont at closed rect region $R: \{(x,y) \in R^2 / |x-x_0| \le a, |y-y_0| \le b\}$ and $f(x,y) \le M$, then ODE $\frac{dy}{dx} = f(x,y)$, $y(x_0) = y_0$ has at least 1 soln in interval $|x - x_0| < h$, $h = \min\{a, b/M\}$.

* **Sufficient Condition**, not necessary.
Like kabhi kabhi $f(x,y)$ continuous nhi hua toh bhi soln exist kar skta hai.

16] **Uniqueness Thm**: ① $f(x,y)$ is continuous at all points in rectangle: $R: |x-x_0| \le a ; |y-y_0| \le b$ & $|f(x,y)| \le M$.

② $f(x,y)$ satisfies the lipschitz condition

$$\boxed{|f(x,y_2) - f(x,y_1)| \le L \cdot |y_2 - y_1|}$$

L= Lipschitz constant

$y_1, y_2, x \in R$.

Then $y' = f(x,y)$, $y(x_0) = y_0$ has a unique soln in $|x-x_0| < h$ &

$h = \min\left\{a, \frac{b}{m}\right\}$

* Sometimes, its diff to check the lipschitz condition. Then if $\frac{\partial f}{\partial y}$ is continuous & bounded in $|x-x_0| \le a$; $|y-y_0| \le b$ $\left|\frac{\partial f}{\partial y}\right| \le N$, then it satisfies the lipshitz condition.

↳ Converse need not be true.

Max of function nikalne ka tareeka:
① Numerators to max karo &
② Denomenator ko min karo.

17] **Picard's Theorem**: When all the above conditions of Uniqueness thm holds true, then

$$y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(x, y_n(x))\, dx$$

$n = 0, 1, 2, 3 \ldots$

then $y_1, y_2, y_3 \ldots$ converge to the unique solution of $\frac{dy}{dx} = f(x,y)$. This is Picard's iterative method.

18] **Solution of Second Order ODE's**:

$$K_2 \frac{d^2y}{dx^2} + K_1 \frac{dy}{dx} + K_0 y = 0$$ K's are const.

$\boxed{y_1 \& y_2}$ are soln

Further $\boxed{c_1 y_1 + c_2 y_2}$ is also the soln.

$c_1 \& c_2 \to$ arbitrary const.

19] **General Soln of n^{th} order ODE**:

n^{th} order will have soln with n arbitrary const.

20] **Linearly Independent**: If $\phi_1, \phi_2, \phi_3 \ldots$ are fns.

& $c_1\phi_1(x) + c_2\phi_2(x) + c_3\phi_3(x) + c_4\phi_4(x) \ldots = 0$

& $c_1 = c_2 = c_3 = \ldots = c_n = 0$ then $\phi_1(x), \phi_2(x) \ldots \phi_n(x)$ are linearly independent.

If there exist atleast 1 non-zero solution, then fns are linearly dependent.

How to find?

ϕ_1, ϕ_2, ϕ_3 are LI if

$$\underbrace{\begin{vmatrix} \phi_1 & \phi_2 & \phi_3 \\ \phi_1' & \phi_2' & \phi_3' \\ \phi_1'' & \phi_2'' & \phi_3'' \end{vmatrix}}_{\text{Wronskian matrix}} \ne 0$$

If $=0$, then L.D. [Proof on pg-68 pdf]

21] **Theorem**: If $y_1 \& y_2$ are soln of $K_2 f'' + K_1 f' + K_0 y = 0$ & $y = c_1 y_1(x) + c_2 y_2(x)$ is the general soln then $y_1 \& y_2$ are L.I.

[Proof on pg-70]

21] **Soln of 2^{nd} Order ODE**: $K_2 \frac{d^2y}{dx^2} + K_1 \frac{dy}{dx} + K_0 y = 0$

& r_1, r_2 are roots then

Case I: If r_1, r_2 are two distinct roots, the soln are

$y_1(x) = e^{r_1 x}$ & $y_2(x) = e^{r_2 x}$ [L.I]

$$\boxed{y = c_1 e^{r_1 x} + c_2 e^{r_2 x}}$$

Case II: If r_1, r_2 are distinct complex no.

$r_1 = p + iq$; $r_2 = p - iq$ then

$y_1 = e^{(p+iq)x}$; $y_2 = e^{(p-iq)x}$

$y = c_1 e^{(p+iq)x} + c_2 e^{(p-iq)x}$

$$\boxed{y = e^{px}[c_3 \cos qx + c_4 \sin qx]}$$

Case III: If $r_1 = r_2$

$$\boxed{y = c_1 x . e^{r_1 x} + c_2 e^{r_1 x}} \quad (c_1 x + c_2) e^{r_1 x}$$

For the solution of ODE to be periodic, there should be only trigo term in soln, exponential & algebraic term must be zero.

22] Homogenous 2^{nd} order LODE: If

$$K_2 \frac{d^2y}{dx^2} + K_1 \frac{dy}{dx} + K_0 y = f(x)$$

where $f(x) = 0$.

Its solution is y_c

23] Non Homogenous: If $f(x) \neq 0$.

Its solution is y_p

Overall solution is $y_{(x)} = y_c(x) + y_p(x)$

24] Method of undetermined coefficients for finding the particular soln:

① Make Auxillary eqn $\frac{d}{dx} = r$.

② Solve it to get roots.

③ You get y_c using previous methods using r_1, r_2.

④ Derivatives, y ke saath jo x wala fn hoga, uske power jitna function banao. like $y_p = A_1 + A_2 x + A_3 x^2 \dots\dots A_{n+1} x^n$

⑤ ~~Then diff it till the order of diff~~ eqn. & substitute in the given ODE.

⑥ Compare coefficients to get value of $A_1, A_2, A_3 \dots$

⑦ $y = y_c + y_p \rightarrow$ General Solution.

Everytime assumption of y_p is not algebraic, it can be expo/trigo too, based of x wala function given in the question.

ex- If $f(x) = e^{7x} + e^{3x}$ then $y_p = Ae^{7x} + Be^{3x}$

$f(x) = x^2 + x + e^{10x}$ then $y_p = a_0 + a_1 x + a_2 x^2 + A \cdot e^{10x}$

※ Sometimes its problematic to take $y_p = Ae^x$, we get no possible soln, then take $y_p = Axe^x \rightarrow$ this gives solution.

or

$y_p = Ax^2e^x \rightarrow$ Depending on $y_c(x)$

when $f(x) = \cos 5x$, then chose $y_p = A\cos 5x + B\sin 5x$.

$f(x) = 12e^{2x}\sin 3x$, then $y_p(x) = e^{2x}[A\cos 3x + B\sin 3x]$

※ Accha y_p choose karna is very imp ☺

25] Method of Variation of Parameters:

for $K_2 \frac{d^2y}{dx^2} + K_1 \frac{dy}{dx} + K_0 y = f(x)$.

let $y_c(x) = Ay_1(x) + By_2(x)$

$y_p(x) = A(x)y_1(x) + B(x)y_2(x)$

$$A(x) = -\int \frac{g(x) \cdot y_2}{W(x)} dx \; ; \; B(x) = \int \frac{g(x) \cdot y_1(x)}{W(x)} dx.$$

$\boxed{g(x) = \frac{f(x)}{K_2}}$ $\quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$

$$y_p(x) = -y_1(x)\int \frac{g(x) \cdot y_2(x)}{W(x)} dx + y_2(x)\int \frac{g(x)\, y_1(x)}{W(x)} dx$$

Procedure - ① First find r, then y_c by normal method.

② Then find $g(x) = f(x)/K_2$, then find y_1, y_2 from y_c.

③ Using y_1, y_2 find W.

④ Find $A(x) = -\int \frac{g(x) y_2(x)}{W(x)} dx$

$B(x) = +\int \frac{g(x) y_1(x)}{W(x)} dx$.

$\therefore y_p = A(x)y_1(x) + B(x)y_2(x)$

then $y = y_c + y_p$.

※ Read Question on pg - 103 - VVImp.

26] Cauchy Diff Eqn:

$$K_2 x^2 \frac{d^2y}{dx^2} + K_1 x \frac{dy}{dx} + K_0 y = f(x) \text{ [Non homogenous]}$$

$$K_2 \frac{d^2y}{dt^2} + (K_1 - K_2)\frac{dy}{dt} + K_0 y = f(\log_e x) \rightarrow \text{Homogenous.}$$

$t = \log_e x$

$y(x) = y_c(\log_e x) + y_p(\log_e x)$

※ Read Question on pg - 110 - VVImp

27] Cauchy Lengendre ODE:

$$K_2(ax+b)^2 \frac{d^2y}{dx^2} + K_1(ax+b)\frac{dy}{dx} + K_0 y = f(x)$$